Crowd Risk Management During Hajj: A Data-Driven Approach for Identifying High-Risk Pilgrim Congestion Zones

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***Abstract*—Hajj, the annual Islamic pilgrimage to Makkah, involves the gathering of millions of pilgrims in a confined area, presenting significant challenges in crowd management and safety. This paper proposes a data-driven approach to manage crowd safety by leveraging machine learning models to predict high risk zones based on factors such as age, health history, environmental conditions, and crowd density. The system integrates these predictions into an interactive dashboard, providing a real- time visualization of risk distribution and enabling authorities to allocate resources more effectively. By proactively addressing potential risks, the proposed solution aims to enhance the safety and experience of pilgrims during Hajj.**

I. Introduction

The Hajj pilgrimage is one of the largest religious gatherings in the world, attracting millions of Muslims to the holy city of Makkah each year. Managing such a massive crowd poses numerous challenges, including ensuring the safety and well-being of pilgrims, preventing overcrowding, and providing timely emergency responses. Factors such as varying health conditions of pilgrims, and the sheer density of people in specific areas make it crucial to implement efficient crowd management strategies. Traditional crowd management approaches often rely on manual monitoring and static contingency plans, which may not adapt well to real-time dynamics. To address these limitations, this project leverages data science techniques to develop a predictive system that identifies high-risk areas and individuals based on key factors. The system combines machine learning with geographic visualization to provide decision-makers with actionable insights for resource allocation and risk mitigation by integrating advanced data analytics and real time visualization. This project aims to transform the way crowd management is conducted during Hajj, ensuring a safer and more organized pilgrimage experience.

II. Literature Review

A study by Ghazzawi and Al-Bahr (2023) explores the ongoing issue of traffic congestion during the Hajj season, where nearly 250,000 vehicles contribute to delays, road accidents, and increased pollution, all of which have significant impacts on public health. The research focuses on how AI and advanced communication technologies can be used to improve traffic management in Makkah, particularly through smart cameras and real-time data analytics. These tools aim to predict and reduce congestion by suggesting alternative routes and facilitating smoother vehicle flow. The researchers highlight various innovative methods for addressing the unique challenges of Hajj and Umrah, emphasizing AI applications like image processing, machine learning, and computer vision. By leveraging the existing network of surveillance cameras in the holy sites, the proposed system can track and predict traffic patterns. For example, smart cameras use algorithms to process images and detect crowding, enabling timely interventions, such as switching entry lanes to exit lanes to ease traffic. Additionally, recent research like “The Hajj Lab” introduces a multi-layered approach to crowd management, combining AI and real-time data with advanced technologies like stratospheric airships. These airships offer aerial surveillance, providing a broader view of crowd dynamics across the holy sites. The real-time data collected is analyzed with AI to predict crowd movements and identify potential bottlenecks, enabling proactive measures. By using advanced algorithms, these systems help monitor crowd behavior and improve safety measures during Hajj. The study also reviews past research on crowd and traffic management, showing that a single solution is often insufficient for the complex challenges of Hajj. Instead, it advocates integrating multiple modern technologies to create a more comprehensive and effective system. These findings underline the importance of adopting diverse strategies to ensure smoother mobility and enhanced safety during the pilgrimage.

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III. Data collection

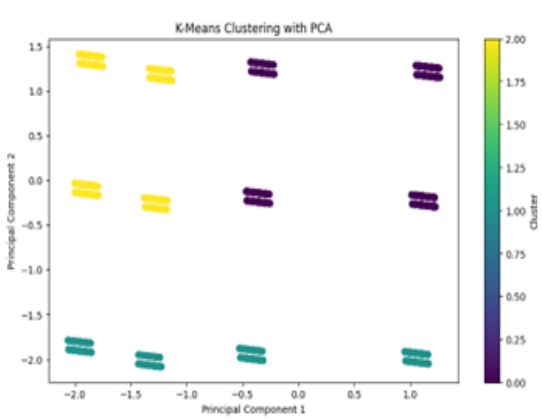
Pilgrim Demographics: Age, health conditions, prior medical history. - Crowd Density Metrics: Real-time foot traffic patterns. - Environmental Factors: Temperature, humidity, and air quality. - Performance Metrics: - Model Accuracy: Prediction success rate. - Classification Precision: Identifying high-risk individuals with minimal false positives. - Efficiency: Reduction in response time to emergencies.

IV. Methodology

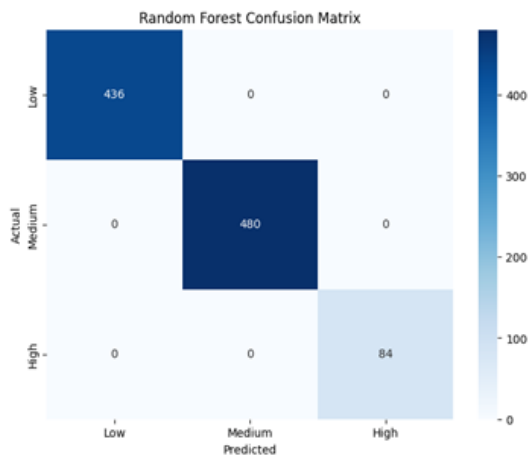
algorithm	objective	why this algorithm	Practical example
Random forest classifier	Accurately classifying cases (Risk/No Risk) and minimizing errors.	Accurate for categorized data; effectively reduces overfitting.	Identifying high-risk pilgrims based on factors such as age and health status.
Neural Network	Understanding complex data patterns to enhance predictions.	Great for large, complex datasets with non-linear relationships.	Predicting health risks using factors like history, density, and age.
Clustering(k-means)	Grouping pilgrims to identify high-risk individuals.	Helps uncover hidden patterns and identify clusters in data.	Identifying high-risk areas for efficient resource allocation.
Adm optimizer	Accelerating training and ensuring stable results.	Efficient for training neural networks and managing large datasets.	Optimizing the neural network for better accuracy and stability faster.
Sparse categorical crossentropy	Assessing the model and enhancing classification accuracy.	Ideal for multi-class classification and accuracy measurement.	Evaluating the model's accuracy in classifying health risk levels for pilgrims.

V. Result and discussion

a) - Based on the results, the analysis indicates that dividing the pilgrims into three clusters is the optimal choice, as the clusters reflect clear distinctions between the groups. This segmentation reveals unique patterns Cluster 1: Elderly pilgrims with poor health conditions and crowded routes, requiring intensive medical support. Cluster 2: Younger pilgrims with good health conditions and less congested routes, making them easier to manage. Cluster 3: Middle-aged pilgrims with stable health conditions and moderately busy routes. This analysis enables a more efficient allocation of resources: enhancing medical services for high-risk groups, improving crowd management in congested routes, and distributing resources based on the specific needs of each cluster. These insights transform the data into actionable strategies, fostering a safer and smoother pilgrimage experience.



b) The Random Forest model achieved a perfect accuracy of 100%, scoring 1.00 in all performance metrics (Precision, Recall, and F1-Score) for all categories: "Low," "Medium," and "High." The confusion matrix showed a complete match between predictions and actual values, with no classification errors. Reasons for High Performance: Data Quality: The well-organized and clear data enabled the model to easily identify patterns. Features Used: Features like age, health condition, and route were highly correlated with the risk level, contributing to precise classification. Model Characteristics: Random Forest is robust in handling diverse data and minimizes overfitting, especially with carefully chosen parameters (e.g., n_estimators=100 and max_depth=10). Balanced Data: The evenly distributed classes (Low, Medium, High) allowed the model to learn effectively without bias. Conclusion: The model delivered exceptional performance, reflecting the quality of the data and the model's strength. However, validating its reliability with real-world, representative data is essential.



c) We use deep learning also but achieve less accuracy than machine learning why is the Neural Network's Performance Lower than Random Forest? Small Data Size: Neural networks require large datasets to perform well, while Random Forest works efficiently with smaller datasets. Simple Relationships: Random Forest is well - suited for data with clear and straightforward patterns, whereas neural networks are designed for complex, non-linear relationships. Overfitting: The neural network may have overfit the data due to its limited size. Time and Complexity: Random Forest is faster and simpler, while neural networks are more complex and take longer to train. So Random Forest is a better fit for your data because it is structured and clear, while neural networks require larger datasets and more complex patterns to outperform.

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Random Forest Predictions:
[0 1 1 0 2 0 0 1 1 2 0 1 0 0 1 0 0 1 1 2 1]
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
Deep Learning Predictions:
[0 1 1 0 2 0 1 1 1 2 0 1 0 0 1 0 0 1 1 2 1]
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d) The developed dashboard provides a real-time visualization of hazardous congestion points on roads leading to Arafat. Using a **Folium-based interactive map**, it marks dangerous locations with red indicators, allowing users to identify high-risk areas quickly. The dashboard integrates synthetic data to simulate real-world traffic conditions and potential hazards. Additionally, it offers a **user-friendly interface** that simplifies navigation and enhances decision- making by displaying congestion intensity and hazard details in an intuitive manner. The system's accuracy depends on model predictions, ensuring that the insights provided are reliable for improving traffic safety during Hajj and Umrah.

VI. Conclusion and Recommendations

Machine learning significantly enhances Hajj crowd management by predicting risk zones and enabling real-time safety interventions. The findings highlight the importance of adaptive AI-driven decision-making. Future improvements should integrate IoT-based monitoring and AI-enhanced simulations to further optimize crowd control strategies.

References

[1] Ghazzawi, F., & Al-Bahr, M. (2023). Reducing traffic congestion in Makkah during Hajj through AI.

[2] Alshahrani, A. (2024). The Hajj Lab: AI Analytics for Crowd Management